Data Science Course - MiniProject II - Students' Performance

Matin Moradi

Åbo Akademi University
Matin.Moradi@abo.fi

Introduction

Have you ever wondered whether it is possible to predict a student's final grade in a course before the semester ends? Or perhaps, as an instructor, you might wish to identify students who are likely to underperform so that timely support can be provided. In many educational settings, large amounts of data are collected from online learning platforms, yet this data is rarely utilized to its full potential.

In this project, we propose the use of machine learning techniques to predict students' final grades based on their learning activities throughout the course. The dataset used in this study was collected from an online course hosted on Moodle. It includes each student's scores on quizzes, mini-projects, and peer reviews, as well as detailed system logs that capture their interactions with the course materials. By analyzing these features, we aim to build predictive models capable of estimating a student's final grade, thereby demonstrating the potential of data-driven methods to enhance educational outcomes.

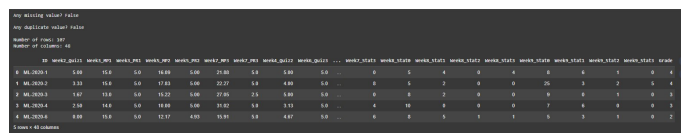
Data Processing

The dataset contains records from 107 students and includes 48 features in total. The attributes are as follows:

```
['ID', 'Week2_Quiz1', 'Week3_MP1',
'Week3_PR1', 'Week5_MP2', 'Week5_PR2',
'Week7_MP3', 'Week7_PR3', 'Week4_Quiz2',
'Week6_Quiz3', 'Week8_Total',
'Week1_Stat0', 'Week1_Stat1',
'Week1_Stat2', 'Week1_Stat3',
'Week2_Stat0', 'Week2_Stat1',
'Week2_Stat2', 'Week2_Stat3',
'Week3_Stat0', 'Week3_Stat1',
'Week3_Stat2', 'Week3_Stat3',
'Week4_Stat0', 'Week4_Stat1',
'Week4_Stat2', 'Week4_Stat3',
'Week5_Stat0', 'Week5_Stat1',
'Week5_Stat2', 'copyWeek5_Stat3',
'Week6_Stat0', 'Week6_Stat1',
'Week6_Stat2', 'Week6_Stat3',
'Week7_Stat0', 'Week7_Stat1',
'Week7_Stat2', 'Week7_Stat3',
'Week8_Stat0', 'Week8_Stat1',
'Week8_Stat2', 'Week8_Stat3',
'Week9_Stat0', 'Week9_Stat1',
'Week9_Stat2', 'Week9_Stat3', 'Grade']
```

For each of the nine weeks, the dataset includes scores for three main tasks: quizzes, mini-projects, and peer reviews (a total of 27 columns). Additionally, four statistical features are recorded per week, representing course-related, assignment-related, grade-related, and forum-related activity logs. The target label, `Grade`, represents the students' final grades and takes integer values from 0 to 5.

An overview of the dataset is presented in Figure 1.



```
Any missing value? False

Any duplicate value? False

Number of rows: 107
Number of columns: 48
```

	ID	Week2_Quiz1	Week3_MP1	Week3_PR1	Week5_MP2	Week5_PR2	Week7_MP3	Week7_PR3	Week4_Quiz2	Week6_Quiz3	...	Week7_Stat1	Week8_Stat0	Week8_Stat1	Week8_Stat2	Week8_Stat3	Week9_Stat0	Week9_Stat1	Week9_Stat2	Week9_Stat3	Grade
0	ML-2020-1	5.00	15.0	5.0	16.69	5.00	21.83	5.0	5.00	5.0	...	0	5	4	0	4	8	6	1	0	4
1	ML-2020-2	3.33	15.0	5.0	17.83	5.00	22.77	5.0	4.00	5.0	...	0	5	2	0	0	26	3	2	5	4
2	ML-2020-3	1.67	13.0	5.0	15.27	5.00	27.95	2.5	5.00	5.0	...	0	8	2	0	0	9	0	5	0	3
3	ML-2020-4	2.50	14.0	5.0	18.00	5.00	39.02	5.0	3.33	5.0	...	4	10	0	0	0	7	0	0	0	3
4	ML-2020-6	8.00	15.0	5.0	12.17	4.93	15.91	5.0	4.67	5.0	...	6	8	5	1	1	5	3	1	0	2

5 rows × 48 columns

Figure 1: Dataset Overview

As shown in the figure, there are no missing values or duplicate rows in the dataset. However, one major challenge arises: the number of features is relatively large compared to the number of data points, which can lead to overfitting and reduced model generalization. With only 107 samples, the dataset size is considerably smaller than typical machine learning datasets, which often include thousands or even millions of records.

To mitigate this issue, we examined feature correlations in order to identify and remove redundant or uninformative attributes. Figure 2 presents the correlation matrix for all features. It can be observed that one feature column contains constant values (an empty row in the matrix), which should be removed. Moreover, task-related scores show the highest correlation with the final grade, whereas the statistical features (`Stat` columns) exhibit weak correlations and can thus be excluded to simplify the model. Moreover, since the `ID` column is not numerical and does not provide useful information for prediction, we drop it before training our models.

After dropping these less informative features, the correlation matrix for the remaining attributes is shown in Figure 3.

Copyright © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

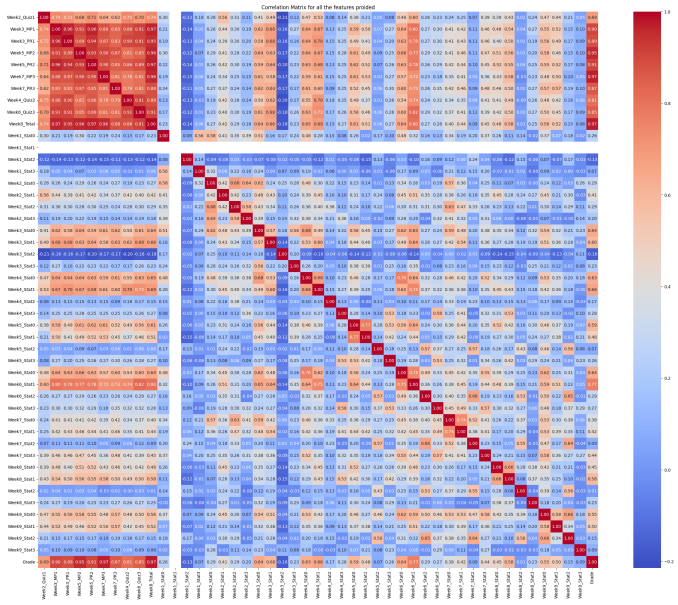


Figure 2: Correlation matrix before feature reduction.

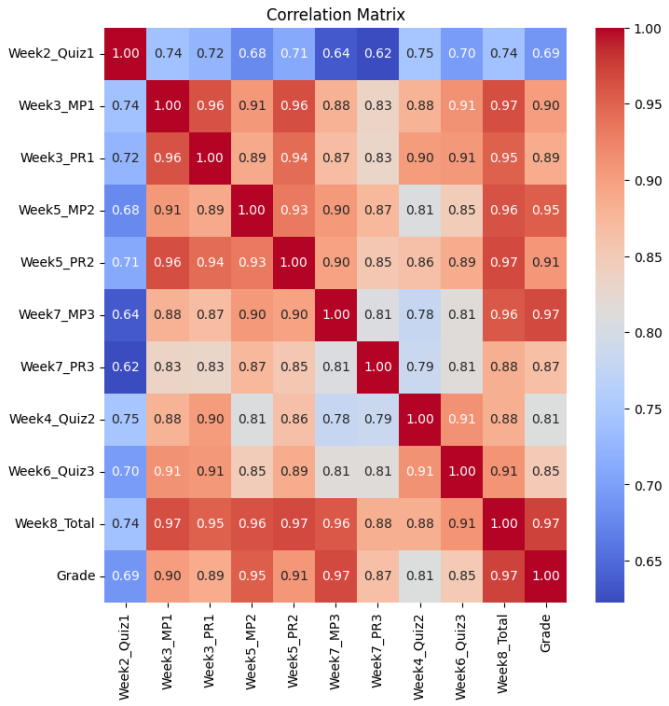


Figure 3: Correlation matrix after removing low-importance features.

Data Split and Model Training

The dataset was divided into two subsets: a training set (80%) used for model learning and a testing set (20%) used for performance evaluation. The goal was to classify students' final grades into six categories ranging from 0 to 5.

Two machine learning models were implemented: **k-Nearest Neighbors (kNN)** and **Random Forest**. The kNN algorithm is a simple yet effective non-parametric model that predicts the class of a sample based on the majority class of its k nearest neighbors in the feature space. In this experiment, k was set to 5 to balance model smoothness and sensitivity to local data variations.

The Random Forest model, on the other hand, is an ensemble method that builds multiple decision trees and combines their outputs through majority voting. This approach reduces variance and improves generalization compared to a single decision tree. To control model complexity and prevent overfitting, the number of trees was limited to 50 and the maximum tree depth was set to 10.

Evaluation and Visualization

The model performance was evaluated using classification accuracy, defined as the ratio of correctly predicted samples to the total number of samples:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

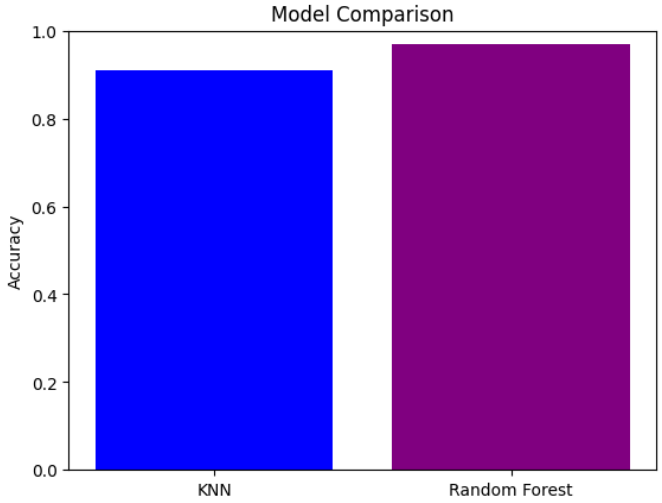


Figure 4: Comparison between kNN and Random Forest Performance

The **k-Nearest Neighbors (kNN)** model achieved an accuracy of 93.94%, while the **Random Forest** model reached 96.97%. As shown in Figure 4, the Random Forest performed slightly better, likely due to its ensemble nature, which reduces variance and improves generalization. In contrast, kNN is a simpler, instance-based learner that relies solely on local data points and does not explicitly learn model parameters. The kNN model performed worse when using a higher value of k, as larger neighborhoods tend to

oversmooth class boundaries and ignore finer patterns in the data.

A visualization of the testing data is presented in Figure 5, where the actual (ground-truth) grades are compared with the predicted grades for each student. The proximity of the red and blue points indicates how closely the model's predictions align with the true outcomes.

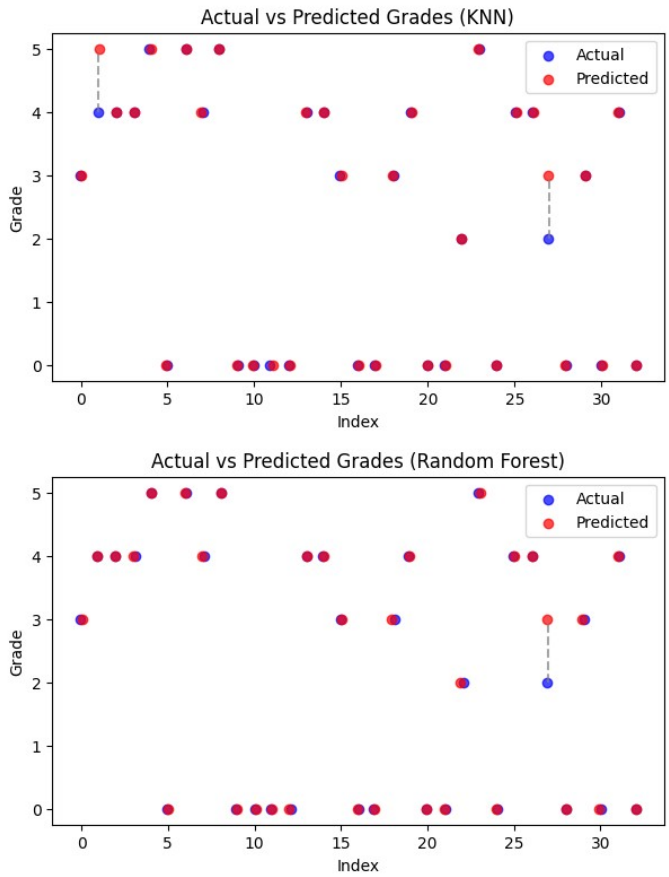


Figure 5: Actual vs. Predicted Grades on Test Data

Important Features

The top ten most influential features identified by the Random Forest model are shown in Figure 6. Among them, `Week8_Total`, `Week7_MP3`, `Week5_MP2`, and `Week3_MP1` stand out as the most important predictors of the final grade. This indicates that students' performance in mini projects, particularly in the later weeks, plays a crucial role in determining their overall success in the course. Hopefully, this mini project will also achieve a high score!

Conclusion

In this project, I applied machine learning techniques to predict students' final grades based on their performance and activity data from an online course. After data preprocessing and feature selection, I trained and evaluated two models—K-Nearest Neighbors and Random Forest. The

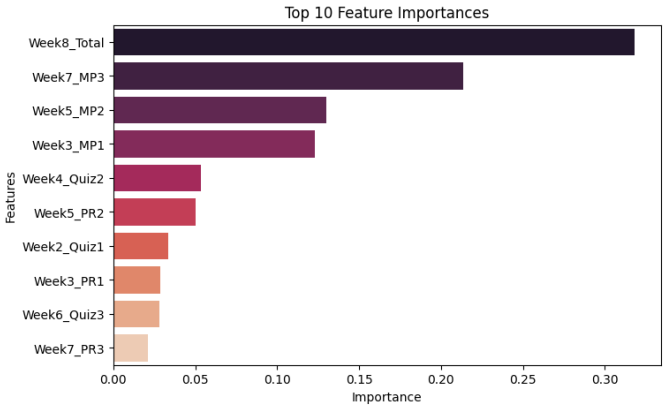


Figure 6: Top 10 important Features

Random Forest model achieved higher accuracy, demonstrating its ability to capture complex relationships within the data.

One major challenge in this study was the large number of features compared to the limited number of samples. Selecting which features to keep was not straightforward, as dropping informative columns could harm performance. By analyzing feature correlations, I identified and removed many features that were less relevant or redundant. Although this approach seemed risky at first, it ultimately led to improved results and better model generalization.

Overall, this study shows that data-driven approaches can effectively predict academic performance and provide valuable insights for instructors to identify students who may need additional support.